Think-A-Dot Instructions

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1.0 Introduction

Start by tilting it on its side . . . then stand it up . . . Now, drop a marble in any of the 3 holes on top . . . Try to make all the spot BLUE . . . or YELLOw . . . by dropping the marble in the holes on the TOP over and over . . . That's the FIRST game you can play with THINK-A-DOT. There are lots more . . .

2.0 Section 1

2.1 OTHER GAMES YOU CAN PLAY WITH THINK-A-DOT

To Play Alone

Start by tilting THINK-A-DOT to the RIGHT so that the pattern looks like this:

```
                Y --- B --- Y

                   B --- B

                Y --- B --- Y
```

Then try to get any of the patterns below in the number of moves indicated below each pattern. (These are only 18 of a possible 128 ptterns. You may work the rest out yourself.)

```
 Y - Y - B   Y - Y - Y   Y - B - Y   B - Y - B   Y - Y - Y   B - B - B

   Y - Y       B - B       B - B       Y - Y       Y - Y       Y - Y

 B - Y - Y   Y - Y - Y   B - B - B   Y - Y - Y   Y - Y - Y   Y - B - Y

    (2)         (3)         (4)         (3)         (7)         (6)

 B - Y - B   B - B - B   Y - B - Y   B - B - B   B - Y - B   B - B - B

   B - B       B - B       Y - Y       B - B       Y - Y       Y - B

 Y - Y - Y   B - B - B   Y - B - Y   Y - B - Y   B - Y - B   B - B - B

    (9)         (6)         (6)         (8)         (7)         (8)

 Y - Y - Y   Y - B - Y   Y - Y - Y   Y - Y - Y   Y - Y - B   Y - Y - Y

   Y - Y       Y - Y       B - Y       Y - B       B - B       B - B

 B - Y - B   B - B - B   B - B - Y   Y - B - B   Y - Y - B   B - Y - B

    (3)         (6)         (3)         (5)         (8)         (7)
```

Next start by tilting THINK-A-DOT to the left so that the pattern looks like this:

```
                B --- Y --- B

                   Y --- Y

                B --- Y --- B
```

Then try to get any of the patterns below in the number of moves indicated below each pattern. (These are 18 of a possible 128 patterns. you may work the rest out for yourself.)

```
 B - B - B   B - B - Y   B - B - B   B - B - B   B - Y - B   B - Y - B

   B - B       Y - Y       B - Y       Y - B       Y - Y       B - B

 Y - B - Y   B - B - Y   B - Y - Y   Y - Y - B   B - Y - B   Y - Y - Y

    (3)         (4)         (3)         (5)         (6)         (6)

 B - B - B   Y - Y - Y   Y - B - Y   B - B - B   Y - B - Y   Y - Y - B

   Y - Y       B - B       Y - Y       Y - B       B - B       Y - Y

 Y - B - Y   Y - Y - Y   Y - B - Y   B - B - B   Y - B - Y   B - Y - Y

    (7)         (8)         (9)         (5)         (7)         (7)

 B - B - Y   B - Y - B   Y - Y - Y   Y - B - Y   B - B - B   B - B - B

   B - B       Y - Y       B - B       B - B       Y - Y       B - B

 Y - B - B   Y - Y - Y   B - Y - B   B - B - B   B - B - B   B - B - B

    (4)         (4)         (6)         (3)         (3)         (7)
```

Answers to all of the above moves are given at the end of this instruction book. Try to predict the new dot pattern that will occur with each drop of the marble. This will help your play.

To Play Against a Friend

As a game, two players try to outguess THINK-A_DOT and each other.

Each player first chooses one of two dot colors as his own. Then they choose a dot pattern as the winning pattern. Two of these might be:

```
      Y - B - Y       Make all the dots in    B - B - B
        B - B         --- this triangle or      B - B
      B - B - B       in this triangle ---    Y - B - Y
                      your color.
```

Or choose any of the patterns given so far or any others you want.

Then set THINK-A-DOT to a starting pattern by either tilting it to one side or the other or shaking it to get a random tarting pattern. Choose who goes first. The second player chooses a side (left or right) of THINK-A-DOT.

The first player may continue his turn of dropping marbles as long as the marble comes out his chosen side.

The other player starts as soon as the marble comes out his side. Then he continues as long as the marble comes out his side.

And so on until one of the players gets the chosen winning pattern ll in his color.

The following discussions are intended for those interested in the Computer Apects of THINK-A-DOT.

3.0 Section II

3.1 THINK-A-DOT AND COMPUTERS

A computer is a tool for working with information. Usually the information takes the form of numbers and words. The computer can memorize information given to it by people. It can then perform various calculations on this information. Results of these calculations can also be remembered by the computer. A person can find out what is in the computer's memory when needed.

Electronic computers are used today to predict the weather and elections, control space shots, and calculate paychecks. These computers can remember large amounts of information (all the records for thousands of employees). They can also operate at very high speeds (calculate and print each employee's paycheck in a fraction of a second). While these large machines are very complicated, the basic ideas used in them are not. Very simple, slow, mechanical computers can be built using the same ideas. These small computers, while not as useful as their larger cousins, are sometimes more fun. THINK-A-DOTIS ONE OF THESE SMALL COMPUTERS. While intended as a game it also illustrates many of the basic idea used in larger machines.

By tilting THINK-A-DOT to the left or right you can put one of two different information (or color) patterns into its memory. Note that when you return THINK-A-DOT to its upright position it retains or "remembers" the pattern you set into it when you tilted it. You can then cause THINK-A-DOT to perform calculations on the memorized information by dropping a marble in at the top. The newly calculated color pattern then replaces the original pattern in THINK-A-DOT's memory. You can observer the information (color pattern) in THINK-A-DOT's memory by means of the eight windows provided. The memory of a large computer might comprize millions of individual memory cells. Each cell, however, could remember only one of two things. For this reason, thay are called binary cells. THINK-A-DOT contains eight binary memory cells, each of which remembers either of its two colors. Another name for these cells is Flip-Flop.

Much of the ability of a large computer comes from using parts which perform "logical" operations. You can easily demonstrate "logical" operations with THINK-A-DOT. Drop the marble into the middle hole. The bottom right-hand dot will change color only when the top-middle AND the middle-right hnd dots are both yellow. This is called a "logical" operation. See if you can discover others in THINK-A-DOT.

4.0 Section III

4.1 THINK-A-DOT OPERATIONS

For those interested in learning how to program THINK_A_DOT the following discussion may be helpful.

(1) Each time a marble hits a flip-flop it changes its color (State) Hence these are complementary flip-flops.

(2) Since the middle and lower flip-flops change color no matter which direction the marble comes from, we can say tht they are connected by "OR" gates. That is to say, a marble coming in from the right OR the left hand path will change (complement) the flip-flop.

(3) The direction the marble goes upon leving flip-flop depend upon its previous color state. For example, when the center top flip-flop is YELLOW the mrble will take the right hand path and move to complement the right hand flip-flop of the center pair. On the other hand, if the top center flip is the BLUE state the marble will move to the left hand path and complement the left hand flip-flop of the center pair. And so on.

NOTE: Not all flip-flops have this same direction routing, of the marble. You will have to experiment with each flip-flop to discover all of the conditions.

(4) Finally, the side on which the mrble comes out depends on the color (state) of the flip-flop it complements in the bottom row of three flip-flops. For example, if the center bottom flip-flop is yellow the marble will come out of the right hand side, if it hits that center flip-flop. And so on . . . You will have to experiment to find the complete set of conditions.

Now, if you have written down all of the above conditions, you can draw a FLOW CHART or diagram for your THINK-A-DOT from which you can program or predict each move. Each flip-flop acts as an information storage element, as well as a logical control element. The marble acts just as an electric current flowing between these elements through OR gates, just as it might in a real electronic computer!

5.0 Section IV

5.1 TEN THINK-A_DOT THEOREMS AND COROLLARIES

The previous description of Section III on the mechanics of opertions can now be applied to prove the following theorems and corollaries for THINK-A-DOT. You may enjoy proving them yourself either by trial or by analysis. Remember that THINK-A-DOT is made up of eight binary (two-state) flip-flops that are flipped and interrogated for their prior state each time a marble hits them.

(1) If a total of 'a', 'b', and 'c' marbles are dropped in the left(L), middle(M), and right(R) holes in the top of THINK-A-DOT, then the new pattern of dots obtained is independent of the order in which they are dropped.

(2) If (8,0,0) or (0,8,0) or (0,0,8) marbles are dropped in the L, M, and R holes respectively, the pattern will be unchanged. ((8,0,0) means, drop eight marbles in the top left hole, zero in the middle hole,

(3) If (2,2,2) marbles are dropped in the L, M, and R holes the pattern will be unchanged. Thi implies that (4,4,4) and (6,6,6) drops will also leave the pattern unchanged.

(4) If the sum of the blue (or yellow) dots in the top and bottom rows, excluding the middle row, is even (or odd) then dropping any number of marbles in L, M, or R will leave thi total even (or odd).

(5) If the starting pattern has an even (or odd) number of blue (or yellow) dots in the top and bottom rows the trget pattern must also have an even (or odd) number of blue (or yellow) dots in these rows for it to be an obtainable pattern. Thus, if the parity agrees, the target pattern can be obtained from the starting pattern. Otherwise, it cnnot.

(6) To obtain a new dot pattern 'P2' from a starting pattern 'P1' proceed as follows:

a - If the top left hand dot in the desired pattern 'P2' does not agree with starting pattern 'P1' drop one marble in L to make it agree. (Otherwise don't drop any marbles in L.)

b - Drop as many marbles as required in M to make the diagonal extending downward from M to the left the same as tht in the new desired pattern 'P2'. (No more than seven marbles will be required.)

c - Drop as many marbles as required in R to make the diagonal extending from R downward to the left the same as 'P2'. (No more than seven marbles will be required.) If 'P2' may be obtained from 'P1' the lower right hand dot will automatically be the correct color after this is done. Otherwise an attempt has been made to transform an 'even' pattern into an 'odd' pattern or vice-versa, which of course, is impossible.

The resulting number of marbles, (a,b,c), dropped in L. M, and R by this technique will be a 'feasible' solution to the problem of transforming P1 into P2 but not necessarily an 'optimum' solution It will also, as a matter of fact, be a 'primitive' solution with these terms defined as follows:

*Parity - merely means that the sum of the dots of one color (often clled 'BITS') is always either even or odd.

A 'feasible' solution, (a,b,c) is a triplet of numbers such that if these numbers of marbles are dropped respectively in L, M, and R in any order an initil pattern P1 will be transformed into a desired pattern P2.

A 'primitive' solution is a feasible solution with no number greater thn 8

An 'optimum' solution i the primitive solution tht makes (a+b+c) a minimum.

(7) All primitive solutions may be obtained as follows:

a - Obtain the firt primitive solution by the method described in 6.

b - Ad (2,2,2) to the numbers (a,b,c) obtained and if ny number obtained is greater than 8 subtract 8 from it.

c - Repeat step 7b on each solution obtained until a prior primitive solution is obtained. Exactly 4 primitive solutions will be obtained.

(8) The optimum solution may now be selected from the primitive solutions in 7 as the one that minimizes the sum of the number of marbles required.

(9) An optimum solution will never require more thn a total of 9 marbles.

(10) All solutions may be obtained by adding multiples of (8,0,0) or (0,8,0) or (0,0,8) to the primitive solutions.

5.2 EXAMPLES OF THE APPLICATION OF THE ABOVE THEOREMS AND CoROLLARIES

Theorem #1

Tilt THINK-A-DOT to the right. A blue diamond will appear in the center with the 4 corners yellow. Put 3 in L, 2 in M and 1 in R. This should transform the pattern to all blue. Now tilt to right again and put marbles in THINK-A-DOT in the order L, M, R, L, M, L. The same pattern should appear.

Theorem #2

Start with any pattern you wish. Then drop 8 marbles in any one hole and see if the same pattern return.

Theorem #3

Drop (2,2,2) marbles in L, M, R and see if the same pattern returns.

Theorem #4

Verify Theorem 4.

Theorem #5

Tilt to right. Drop marbles in any hole and note that the number of blue dots always remains the same.

Theorem #6

Tilt to right. Try to make all dots blue by the method described in Theorem 6. This should take (1,0,7). Now try to obtain all yellow from all blue.

Theorem #7

Note that (1,0,7) + (2,2,2) = (3,2,9) = (3,2,1).Thus, (3,2,1) will also transform the original pattern to all blue. Find all primitive solutions.

Theorem #8

Verify that (3,2,1) is the optimum solution.

Theorem #9

If (1,7,5) is a solution what is the optimum solution? Note that the maximum of 9 marbles is required for the optimum solution.

6.0 HOW THINK-A-DOT CAN BE USED AS A COMPUTER To COUNT, ADD AND SUBTRACT

We will now be able to let you prove to yourself that THINK-A-DOT is really a binary digital computer by explaining how it can count, add and subtract. In order to do this we will first examine number systems so that you may learn how to communicate with your THINK_A_DOT.

This discussion is not intended to be complete. We expect you to work out additional programs and techniques for using THINK-A-DOT.

6.1 Number SYstems

All digital computers operate in a number system known as the Binary System. The primary reasons for the use of such a system are its simplicity and the fact that it can be so easily mechanized. There are only two digits, '0' and '1' (called bits) in the system. Common examples of this simplicity and use are:

```
An electricl switch which is either on or off; a particular
location on a computer punch card which either has a hole or
does not; or a mechanical flip-flop which either shows blue or
yellow as in THINK-A-DOT.
```

In each case, the device has just two stable, mutually exclusive conditions. The digit value 0 is assigned to one of these conditions and 1 to the other.

In ordinary use, binary numbers would be quite cumbersome, since about 3-1/3 binary bits are required to represent a single decimal digit. THis represents no problem to the mammoth electronic computers because of their high speedsand large number of electrical components.

Before going into these other number systems in great detail, we should review the inner workings of the *decimal system* (base 10). The central feature of the deciaml or arabic number system is the fact that a given digit can have more or less value depending on where it is written. For instance, the symbol 2 may mean 2.0, 200., or .0002 depending upon where it is written with respect to the decimal point. This is so familiar to us as to be overlooked in everyday use of decimal numbers. For example, what e really mean by 2756 is, of course:

```
2000 + 700 + 50 + 6 = 2756

or 2 x 10^3  +  7 x 10 ^ 2  +  5 x 10^1  +  6x 10 = (2756)
                                                         10
```

This is the basis for calling the decimal system a 'base 10' system.

In the *binary system* (base 2) the number (1101) base 2 would actually mean:

```
1 x 2^3  +  1 x 2^2  +  1 x 2^1  +  1 x 2^0

or  1 x 8    +  1 x 4    +  1 x 2    +  1 x 1    = (13) base 10
```

The subscript at the end of the parenthesis indicates the base of the number system. Thus ( ) base 2 means the base 2 or the binary system. Or ( ) base 10 means the base 10 or decimal system.

Similarly,in an *octal number system* where there are 8 characters (the digits 0 thru 7) a number (6573)8 would mean:

```
(6573)8 = 6 x 8^3 +  5 x 8^2  +  7 x 8^1  +  3 x 8^0

or 6 x 512  +  5 x 64  +  7 x 8  +  3 x 1 = (3451) base 10
```

Thus any number system may be converted to our decimal system by using each number position as the multiplier for the base number raised to the power corresponding to that position.

Programmers for computers often use the octal base system for the simple reason that the conversion from binary to octal can be carried out mentally, and only a third as many digits are required to carry the same information. All that is necessary to convert from binary to octal is to group the binary bits in groups of three's and write down the value of each group taken as an integer. For instance (11111010110)2 would be written as (011,111,010,110)2 = (3726)8. Here 011=3, 111=7, 010=2 and 110=6 using the binary conversion system discussed above.

The simplicity of conversion could be accomplished using any base which is a power of 2. For instance, a base of 4 could have been used, however, it would not eliminate the cumbersomeness of the large binary numbers. A base such as 16 or 32 could also have been used here, but here it would have been necessary to invent new characters for digits over 9. SO the octal base 8 system is often used for input-output information in computers.

Use of the Octal System for THINK-A-DOT and know it can count from 0 to 128

THINK-A-DOT is a binary digital computer that is most easily programmed with the octal base number system for input/output information. This is because there are 4 flip-flops in each vertical line. If we drop a marble in the right hand hole 8 times, the three right hand flip-flops will form all possible color combinations without duplication. To see this, first tilt THINK-A-DOT to the left and then of course set it upright. Then looking at the three right hand most colors -

```
From Top to Bottom they are              Blue    Yellow  Blue
Now, drop a marble in R and they become       Yellow  Yellow  Yellow
The seocnd drop gives                         Blue    BLue    Blue
The third drop gives                          Yellow  Blue    Yellow
THe fourth drop gives                         Blue    Yellow  Yellow
The fifth drop gives                          Yellow  Yellow  Blue
The sixth drop gives                          Blue    Blue    Yellow
The seventh drop gives                        Yellow  BLue    Blue
The eighth drop gives                         Blue    Yellow  Blue
```

Now we will assign values to the colors, 1 for blue and 0 for yellow and re-write the sequence as follows:

```
0 -- 101      4 -- 100        8 -- 101
1 -- 000      5 -- 001
2 -- 111      6 -- 110
3 -- 010      7 -- 011
```

While these numbers are not the normal binary-decimal translation, the three bit groups are *mutually exclusive* an can therefore be used as a code to define the equivalence of the color patterns to the digits of the number system. With this relationship as a basis we can now build a numbering system around THINK-A-DOT which can be used for counting, adding and subtracting octal numbers.

If we number the flip-flops as follows:

```
              7 --- 4 --- 1

                 5 --- 2

              8 --- 6 --- 3
```

Then the starting pattern (tilt left) could be shown in our binary code as:

```
              1 0 1 0 0 0 1 1
```

where the left hand bit corresponds to the #1 flip-flop and the right hand bit to the #8 flip-flop (blue=1 and yellow=0)

Since there are 8 flip-flops there should be 256 posible color combinations. However, according to Theorem 4 (see Section IV) only half of these patterns are attainable from the starting position. Thus, the next stepo will be to develop a table of binary code to represent the numbers from 0 to 7 will be s previously developed. To form the code for 8 however, an eighth marble will have to be dropped in R and an additional one will have to be dropped in M (the carry).

The table is as follows:

dec	oct	L	M	R	123 456 78	dec	oct	L	M	R	123 456 78
0	0	0	0	0	101 000 11	64	100	1	0:8	0:8	101 011 01
1	1	0	0	1	000 000 11	65	101	1	0:8	1	000 011 01
2	2	0	0	2	111 000 11	66	102	1	0:8	2	111 011 01
3	3	0	0	3	010 000 11	67	103	1	0:8	3	010 011 01
4	4	0	0	4	100 001 11	68	104	1	0:8	4	100 010 01
5	5	0	0	5	001 001 11	69	105	1	0:8	5	001 010 01
6	6	0	0	6	110 001 11	70	106	1	0:8	6	110 010 01
7	7	0	0	7	011 001 11	71	107	1	0:8	7	011 010 01
8	10	0	1	0:8	110 100 11	72	110	1	1	0:8	110 111 01
9	11	0	1	1	011 100 11	73	111	1	1	1	011 111 01
10	12	0	1	2	101 101 11	74	112	1	1	2	101 110 01
11	13	0	1	3	000 101 11	75	113	1	1	3	000 110 01
12	14	0	1	4	111 101 11	76	114	1	1	4	111 110 01
13	15	0	1	5	010 101 11	77	115	1	1	5	010 110 01
14	16	0	1	6	100 100 11	78	116	1	1	6	100 111 01
15	17	0	1	7	001 100 11	79	117	1	1	7	001 111 01
16	20	0	2	0:8	110 011 11	80	120	1	2	0:8	110 001 00
17	21	0	2	1	011 011 11	81	121	1	2	1	011 001 00
18	22	0	2	2	101 010 11	82	122	1	2	2	101 000 00
19	23	0	2	3	000 010 11	83	123	1	2	3	000 000 00
20	24	0	2	4	111 010 11	84	124	1	2	4	111 000 00
21	25	0	2	5	010 010 11	85	125	1	2	5	010 000 00
22	26	0	2	6	100 011 11	86	126	1	2	6	100 001 00
23	27	0	2	7	001 011 11	87	127	1	2	7	001 001 00
24	30	0	3	0:8	100 110 11	88	130	1	3	0:8	100 100 00
25	31	0	3	1	001 110 11	89	131	1	3	1	001 100 00
26	32	0	3	2	110 110 11	90	132	1	3	2	110 100 00
27	33	0	3	3	011 110 11	91	133	1	3	3	011 100 00
28	34	0	3	4	101 111 11	92	134	1	3	4	101 101 00
29	35	0	3	5	000 111 11	93	135	1	3	5	000 101 00
30	36	0	3	6	111 111 11	94	136	1	3	6	111 101 00
31	37	0	3	7	010 111 11	95	137	1	3	7	010 101 00
32	40	0	4	0:8	100 000 10	96	140	1	4	0:8	100 011 00
33	41	0	4	1	001 000 10	97	141	1	4	1	001 011 00
34	42	0	4	2	110 000 10	98	142	1	4	2	110 011 00
35	43	0	4	3	011 000 10	99	143	1	4	3	011 011 00
36	44	0	4	4	101 001 10	100	144	1	4	4	101 010 00
37	45	0	4	5	000 001 10	101	145	1	4	5	000 010 00
38	46	0	4	6	111 001 10	102	146	1	4	6	111 010 00
39	47	0	4	7	010 001 10	103	147	1	4	7	010 010 00
40	50	0	5	0:8	111 100 10	104	150	1	5	0:8	111 111 00
41	51	0	5	1	010 100 10	105	151	1	5	1	010 111 00
42	52	0	5	2	100 101 10	106	152	1	5	2	100 110 00
43	53	0	5	3	001 101 10	107	153	1	5	3	001 110 00
44	54	0	5	4	110 101 10	108	154	1	5	4	110 110 00
45	55	0	5	5	011 101 10	109	155	1	5	5	011 110 00
46	56	0	5	6	101 100 10	110	156	1	5	6	101 111 00
47	57	0	5	7	000 100 10	111	157	1	5	7	000 111 00
48	60	0	6	0:8	111 011 10	112	160	1	6	0:8	111 001 01
49	61	0	6	1	010 011 10	113	161	1	6	1	010 001 01
50	62	0	6	2	100 010 10	114	162	1	6	2	100 000 01
51	63	0	6	3	001 010 10	115	163	1	6	3	001 000 01
52	64	0	6	4	110 010 10	116	164	1	6	4	110 000 01
53	65	0	6	5	011 010 10	117	165	1	6	5	011 000 01
54	66	0	6	6	101 011 10	118	166	1	6	6	101 001 01
55	67	0	6	7	000 011 10	119	167	1	6	7	000 001 01
56	70	0	7	0:8	101 110 10	120	170	1	7	0:8	101 100 01
57	71	0	7	6	000 110 10	121	171	1	7	6	000 100 01
58	72	0	7	2	111 110 10	122	172	1	7	2	111 100 01
59	73	0	7	3	010 110 10	123	173	1	7	3	010 100 01
60	74	0	7	4	100 111 10	124	174	1	7	4	100 101 01
61	75	0	7	5	001 111 10	125	175	1	7	5	001 101 01
62	76	0	7	6	110 111 10	126	176	1	7	6	110 101 01
63	77	0	7	7	011 111 10	127	177	1	7	7	011 101 01
						128	200	2	0:8	0:8	101 011 10
dec	oct	L	M	R	123 456 78	dec	oct	L	M	R	123 456 78

```
*Note:The eighth marble drop is marked 0 again because the pattern for
that hole repeats itself after every eight drops. This allow the
number of marbles dropped in each hole to equal the corresponding
octal number.
```

Try making some of the numbers listed in the Table. For instance, the decimal number (113)10 is represented by the octal number (161)8 which is formed (after first tilting to the left) simply by dropping 1 marble in L, 6 in M, and 1 in R. Keep practicing with the numbers until you understand how to form them. Note that in doing everything above you have been making THINK-A-DOT COUNT.

To make the transformations directly, notice for instance, that (128)10 is actually (2)x64 = 2 x 8^2. The 8^2 is represented by the L hole. So drop (2) marbles in L and the number (128)10 is formed. As another example, the number (119)10 can be written 1 x 64 + 6 x 8 + 7 x 1 = (1) x 8^2 + (6) x 8^1 + (7) x 8^0, so to make this number drop 1 in L, 6 in M, and 7 in R. Generally, the technique is to subtract out the larget power of 8 as many times as possible, then go to the next lower power of 8 and so on. This laborious process is not really necessary since the above conversion table may be used.

Addition with THINK-A-DOT

To add two decimal numbers first look up their octal equivalents in the table and then drop the corresponding summation of marbles in ech hole. If the summation in a given hole requires 8 or more drops, subtract 8 from the number nd drop (carry) 1 in the next higher order hole (always to the left).

As an example to add (45)10 + (63)10

```
                              L  M
                      Carry   1  1

      (45) 10                   (5  5) 8

   +  (63) 10                   (7  7) 8
 -------------               -----------
     (108) 10                (1  5  4) 8
```

To do the example with THINK-A-DOT, enter 55 by dropping 5 in R and 5 in M. Next drop 7 in R and 1 in M since there have been more than 8 marbles dropped in R. Lastly, drop 7 in M ans 1 in L for the carry (since more than 8 marbles have also been dropped in M). Now look in the chart under (108) 10 and you will see the code for the pattern on THINK-A-DOT.

The above example had two carries. The following example has but one carry.

```
                              L  M
                      Carry      1

      (15) 10                   (1  7) 8

   +  (36) 10                   (4  4) 8
 -------------               -----------
      (51) 10                   (6  3) 8
```

This would be done on THINK-A-DOT by entering (17) 8 s 7 marbles in R (after tilting to the left) nd 1 marble in M. Then drop 4 marbles in R and 4 in M, plus one more in M for the carry of 1.

Now look up the pattern (00101010) which you should have on the machine in the above table. The answer should, of course, be an octal (63) 8 or the decimal 51.

Try adding some more numbers. Don't forget to carry 1 everytime more thn 8 marbles have been dropped in any hole.

Subtraction

Subtraction in a computer is carried out by addition of complements. To understand the basis for this operation let us look at a problem of subtraction of decimal numbers.

493,201 - 126,944 = 493,201 + (1,000,000 -126,944) - 1,000,000

The subtraction 1,000,000 - 126,944 can be very simply performed by subtractin each digit from 9, except the low order one, (in this case the right hand 4) which is subtracted from 10. The answer to this subtraction (873,056) is then called the *'s complement*.

Thus 1,000,000 - 126,944 = 873,056

Now adding 493,201

```
+ 873,056
---------
```

gives you 1,366,257

The 1,000,000 is subtracted off simply by deleting the 1 at the beginning, giving the answer 366,257.

An alternate way of thinking of the same process involves what is known as the *nine's complement*. It is formed by subtracting each digit from 9 *including* the low order. After the addition of the complement, the leading 1 (from the 1,000,000 in our example) is deleted and a 1 is added to the units position (because we subtracted the low order digit from 9 instead of 10). This process is sometimes called an "end-around-carry".

In either method, if a larger number is subtracted from a smaller number, a negative difference will result. The negative difference will be in complement form and must be reconverted. In hand computations it is wise to be sure to complement the smaller of the two numbers and avoid this problem.

This may seem to be an exceedingly difficult way of subtracting two numbers. It is used (or the equivalent process in binary) because it is far simpler to form a digit-by-digit complement than to build circuits or devices to "borrow", as is ordinarily done in everyday deciaml arithmetic. This is especially true of binary, where the one's complement is formed simply by changing ones to zeros and zeros to ones, which can be done with great ease electronically.

If the reader would like to read more thoroughly about computer manipulations of binary numbers, may we suggest DIGI-COMP I with its Advanced Instruction Manual. (See end of this Manual).

Octal subtraction can be carried out in a similar manner. The final part of the operation is to form the "sevens complement". Then add in the usual fashion and perform the "end-around-carry". For instance, to subtract (47603) 8 from (56162) 8 first find the sevens complement of the smaller number (47603) 8 which is (30174) 8 and add as follows:

```
     (56162) 8

 +   (30174) 8    [ complement of (47603) 8 ]
 -------------
    (106356) 8
```

Next perform the "end-around-carry" by removing the high order 1 and adding it to the low order position. This will give the answer of (6357) 8. To check the answer simply add the answer to (47603) 8 as follows:

```
     (47603) 8

 +    (6357) 8    [ complement of (47603) 8 ]
 -------------
     (56162) 8
```

Try a few more problems to be sure you know how to do it. New we are prepared to subtract using THINK-A-DOT. To illustrate the process, suppose we are to subtract (67) 10 from (93) 10 giving an answer of (26) 10. First we look up the octal equivalents of 67 and 93 which re (103) 8 and (135) 8 respectively. Then find the sevens complement of the smaller number (103) 8 which is (674) 8. Now enter (135) 8 in THINK-A-DOT by dropping 1 in L, 3 in M and 5 in R. Add to this (674) 8 by dropping 6 in L, 7 in M, and 4 in R. Remember that when you drop more than eight in a given hole, you must "carry" or drop another marble in the next higher order hole. Thus, the next thing to do is to drop another marble in each L and M. The last step is to "end-around-carry". So just drop one marble in R. (We didn't need to remove the high order 1, as it would have been in the fourth position which is non-existent.) To check your answer look up (26) 10 in the table and you will see the binary code for the pattern in your THINK-A-DOT.

As an another exmple, suppose we subtract (6) 10 from (14) 10.

```
      14            (16) 8          (16) 8

      -6            (-6) 8          (+1) 8    (sevens complement)
      --            ----            ----

      (8) 10        (10) 8          (17) 8

                                    (+1) 8    (end-around-carry)
                                    ----

                                    (10) 8    (answer)
```

To do this problem enter (16) 8, add (1) 8 and "end-around-carry:. This may be done simply by dding 1 in R. (In this case it is not wise to carry to M because it would just have to be subtracted out).

The best rule for handling the high order 1 is to just neglect to carry the highest orer bit. A little practice will make this procedure clear.

On Page 15 we said that THINK-A-DOT could count to 128; however, the reader may form the numbers from 129 to 256 by first tilting THINK-A-DOT to the right and then, using a pencil, to hold the upper right hand flip-flop and tilting THINK-A-DOT to the left so that you get 001 000 11. You should now be able to add and subtract with numbers between (129) 10 and (256) 10.

Answers:

```
R=Right, L=Left, M=Middle

Answers to the THINK-A-DOT problems on Page 0. (don't look at them
unless you really have to.)

A=RM             B=MLL            C=MMMM           D=LMR

E=MMMMMRR        F=MMRRRL         G=MMRRRRLMR      H=MMRLLL

I=MLLLLM         J=RRRLLLLL       K=MMMMMRL        L=MLLLLRML

M=RRM            N=MMRRRR         O=MMM            P=MMMMM

Q=MMMRLLLL       R=MLLMMMM

Answers for Page 1:

A=LLM            B=MRRR           C=MMM            D=MMMMM

E=RMRLML         F=MMLLLL         G=MRRMMMM        H=LLLLLRRR

I=MMMLRRRRR      J=MRRRR          K=MMMMMLR        L=LLLRRRR

M=LLRR           N=MMMM           O=MMLLLR         P=LMR

Q=MRR            R=MMMMMLL
```

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End of Think-A-Dot Instructions